

Fourier series

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin (nx) dx$$

Q: $f(x) = \begin{cases} 0, & -\pi < x < -\pi/2 \\ 1, & -\pi/2 < x < \pi/2 \\ 0, & \pi/2 < x < \pi \end{cases}$

sol:

$$a_0 = \frac{1}{\pi} \left(\int_{-\pi}^{-\pi/2} 0 dx + \int_{-\pi/2}^{\pi/2} 1 dx + \int_{\pi/2}^{\pi} 0 dx \right)$$

$$a_0 = \frac{1}{\pi} \left(0 + (x)_{-\pi/2}^{\pi/2} + 0 \right)$$

$$a_0 = \frac{1}{\pi} \left(\frac{\pi}{2} + \frac{\pi}{2} \right) \Rightarrow a_0 = 1$$

$$a_0 = 1$$

$$a_n = \frac{1}{\pi} \left[\int_{-\pi}^{-\pi/2} 0 \cos nx \, dx + \int_{-\pi/2}^{\pi/2} 1 \cos nx \, dx + \int_{\pi/2}^{\pi} 0 \cos nx \, dx \right]$$

$$a_n = \frac{1}{\pi} \left[0 + \left[\frac{\sin(nx)}{n} \right]_{-\pi/2}^{\pi/2} + 0 \right]$$

$$a_n = \frac{1}{\pi} \left[\sin\left(\frac{n\pi}{2}\right) - \sin\left(-\frac{n\pi}{2}\right) \right]$$

$$a_n = \frac{1}{\pi} 2 \sin\left(\frac{n\pi}{2}\right)$$

$$n=1 \quad a_1 = \frac{1}{\pi} 2 \sin\left(\frac{\pi}{2}\right) \quad a_1 = \frac{2}{\pi}$$

$$n=2 \quad a_2 = 0$$

$$n=3 \quad a_3 = -\frac{2}{3\pi}$$

$$n=4 \quad a_4 = 0$$

$$n=5 \quad a_5 = \frac{2}{5\pi}$$

$$b_n = \frac{1}{\pi} \left(\int_{-\pi}^{\pi} f(x) \sin(nx) \, dx \right)$$

$$b_n = \frac{1}{\pi} \left[\int_{-\pi/2}^{\pi/2} 0 \sin(nx) dx + \int_{-\pi/2}^{\pi/2} 1 \sin(nx) dx + \int_{\pi/2}^{\pi} 0 \sin(nx) dx \right]$$

$$b_n = \frac{1}{\pi} \left[0 + \left[-\frac{\cos nx}{n} \right]_{-\pi/2}^{\pi/2} + 0 \right]$$

$$b_n = \frac{-1}{n\pi} \left[\cos\left(\frac{n\pi}{2}\right) - \cos\left(-\frac{n\pi}{2}\right) \right]$$

$$b_n = \frac{0}{n\pi} \left(-\cos\left(\frac{n\pi}{2}\right) + \cos\left(\frac{-n\pi}{2}\right) \right)$$

$$n=1 \quad b_n = \frac{1}{\pi} \left(-\cos\left(\frac{\pi}{2}\right) + \cos\left(-\frac{\pi}{2}\right) \right)$$

$$b_n = 0$$

$$f(x) = \frac{1}{2} (1) + \frac{2 \cos x}{\pi} - \frac{2 \cos 3x}{3\pi} + \frac{2 \cos 5x}{5\pi}$$

2.

Ex 8.1

evaluate f_0

Q7 $f(x) = \begin{cases} 0 & -\pi \leq x < 0 \\ x & 0 \leq x \leq \pi \end{cases}$

Soln $a_0 = \frac{1}{\pi} \left[\int_{-\pi}^0 0 dx + \int_0^{\pi} x dx \right]$

$$a_0 = \frac{1}{\pi} \left[0 + \left[\frac{x^2}{2} \right]_0^{\pi} \right]$$

$$a_0 = \frac{1}{\pi} \left(\frac{\pi^2}{2} - 0 \right), \quad a_0 = \frac{\pi}{2}$$

$$a_n = \frac{1}{\pi} \left[\int_{-\pi}^0 0 \cos nx dx + \int_0^{\pi} x \cos nx dx \right]$$

$$a_n = \frac{1}{\pi} \left[0 + \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi} \right]$$

$$a_n = \frac{1}{\pi} \left[\frac{\pi \sin(n\pi)}{n} + \frac{\cos n(\pi)}{n^2} - 0 - \frac{\cos(0)}{n^2} \right]$$

$$a_n = \frac{1}{\pi} \left[\frac{\pi \sin(n\pi)}{n} + \frac{\cos n\pi}{n^2} - \frac{1}{n^2} \right]$$

$$a_n = \frac{\sin(n\pi)}{n} + \frac{\cos(n\pi)}{\pi n^2} - \frac{1}{\pi n^2}$$

$$n=1 \Rightarrow a_1 = 0 + \frac{-1}{\pi(1)^2} - \frac{1}{\pi(1)^2} = \frac{-1}{\pi} - \frac{1}{\pi} = \frac{-2}{\pi}$$

$$n=2 \Rightarrow a_2 = 0 + \frac{1}{\pi(2)^2} - \frac{1}{\pi(2)^2} = \frac{1}{4\pi} - \frac{1}{4\pi} = 0$$

$$n=3 \Rightarrow a_3 = 0 + \frac{-1}{9\pi} - \frac{1}{9\pi} = \frac{-2}{9\pi}$$

$$n=4 \Rightarrow a_4 = 0 + \frac{1}{16\pi} - \frac{1}{16\pi} = 0$$

$$n=5 \Rightarrow a_5 = 0 + \frac{-1}{25\pi} - \frac{1}{25\pi} = \frac{-2}{25\pi}$$

$$b_n = \frac{1}{\pi} \left[\int_{-\pi}^0 \sin nx \, dx + \int_0^{\pi} x \sin nx \, dx \right]$$

$$b_n = \frac{1}{\pi} \left[0 + \left[-\frac{x \cos nx}{n} + \frac{\sin nx}{n^2} \right]_0^{\pi} \right] x \begin{matrix} + \sin nx \\ - \cos nx \\ n \end{matrix}$$

$$b_n = \frac{1}{n\pi} \left[\frac{\sin nx}{n} - x \cos nx \right]_0^{\pi} \begin{matrix} 0 \\ + \frac{\sin nx}{n^2} \\ - \frac{\cos nx}{n} \end{matrix}$$

$$b_n = \frac{1}{n\pi} \left[\frac{\sin n\pi}{n} - \pi \cos n\pi - \frac{\sin(0)}{n} + 0 \cos(0) \right]$$

$$b_n = \frac{\sin n\pi}{\pi n^2} - \frac{\cos n\pi}{n} - 0 + 0$$

$$b_n = \frac{\sin(n\pi)}{\pi n^2} - \frac{\cos(n\pi)}{n}$$

$$n=1 \Rightarrow b_1 = 0 - \frac{\cos(\pi)}{1} = -(-1) = 1$$

$$n=2 \Rightarrow b_2 = 0 - \frac{\cos(2\pi)}{2} = -\frac{1}{2}$$

$$n=3 \Rightarrow b_3 = 0 - \frac{\cos(3\pi)}{3} = -\frac{(-1)}{3} = \frac{1}{3}$$

$$n=4 \Rightarrow b_4 = 0 - \frac{\cos(4\pi)}{4} = -\frac{1}{4}$$

$$n=5 \Rightarrow b_5 = 0 - \frac{\cos(5\pi)}{5} = -\frac{(-1)}{5} = \frac{1}{5}$$

$$f(x) = \frac{1}{2} \left(\frac{\pi}{2} \right) + \left(\frac{-2}{\pi} \cos(1 \cdot x) + 0 + \frac{-2}{4\pi} \cos(3x) - \frac{2}{25\pi} \cos(5x) \right. \\ \left. + \sin(x) - \frac{1}{2} \sin(2x) + \frac{1}{3} \sin(3x) - \frac{1}{4} \sin(4x) + \frac{1}{5} \sin(5x) \right)$$

$$f(x) = \frac{\pi}{4} \left(\frac{-2}{\pi} \cos(x) + \frac{-2}{4\pi} \cos(3x) - \frac{2}{25\pi} \cos(5x) + \sin(x) - \frac{1}{2} \sin(2x) \right. \\ \left. + \frac{1}{3} \sin(3x) - \frac{1}{4} \sin(4x) + \frac{1}{5} \sin(5x) \right)$$

(Q4) $f(x) = x^2$, $-\pi \leq x \leq \pi$

Sol $f(x) = \frac{1}{\pi} a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx$$

$$a_0 = \frac{1}{\pi} \left[\frac{x^3}{3} \right]_{-\pi}^{\pi}, a_0 = \frac{1}{3\pi} (\pi^3 - (-\pi)^3)$$

$$a_0 = \frac{1}{3\pi} \cdot 2\pi^3, a_0 = \frac{2\pi^3}{3\pi}, a_0 = \frac{2\pi^2}{3}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos nx dx$$

$$a_n = \frac{1}{\pi} \left[\frac{x^2 \sin(nx)}{n} + \frac{2x \cos(nx)}{n^2} - \frac{2 \sin(nx)}{n^3} \right]_{-\pi}^{\pi}$$

x^2	$\times$	$\cos nx$
$2x$	$\times$	$-\frac{\sin nx}{n}$
2	$\times$	$-\frac{\cos nx}{n^2}$
0	$\times$	$-\frac{\sin nx}{n^3}$

Ans: $\frac{2\pi^2}{3}$

$$a_n = \frac{1}{n\pi} \left[x^2 \sin(nx) + \frac{2x \cos(nx)}{n} - \frac{2 \sin(nx)}{n^2} \right]_{-\pi}^{\pi}$$

$$a_n = \frac{1}{n\pi} \left[\pi^2 \sin(n\pi) + \frac{2\pi \cos(n\pi)}{n} - \frac{2 \sin(n\pi)}{n^2} \right]$$

$$- \left[(-\pi)^2 \sin(n(-\pi)) + \frac{2(-\pi) \cos(-n\pi)}{n} - \frac{2 \sin(-n\pi)}{n^2} \right]$$

$$a_n = \frac{1}{n\pi} \left[0 + \frac{2\pi \cos(n\pi)}{n} - 0 - \left(0 - \frac{2\pi \cos(-n\pi)}{n} + 0 \right) \right]$$

$$a_n = \frac{1}{n\pi} \left[\frac{2\pi \cos(n\pi)}{n} + \frac{2\pi \cos(-n\pi)}{n} \right]$$

$$a_n = \frac{2\pi}{n^2\pi} \left[\cos(n\pi) + \cos(-n\pi) \right]$$

$$a_n = \frac{2}{n^2} \left[\cos(n\pi) + \cos(-n\pi) \right]$$

$$n=1 \Rightarrow a_1 = 2 \left[-1 - 1 \right], \quad a_1 = 2(-2) \Rightarrow a_1 = -4$$

$$n=2 \Rightarrow a_2 = \frac{2}{4} \left[1 + 1 \right], \quad a_2 = \frac{2}{2}, \quad a_2 = 1$$

$$n=3 \Rightarrow a_3 = \frac{2}{9} (-2), \quad a_3 = -\frac{4}{9}$$

$$n=4 \Rightarrow a_4 = \frac{2}{16} (2), \quad a_4 = \frac{1}{4}$$

$$n=5 \Rightarrow a_5 = \frac{2}{25} (-2), \quad a_5 = -\frac{4}{25}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \sin(nx) dx$$

$$b_n = \frac{1}{\pi} \left[x^2 \left(-\frac{\cos nx}{n} \right) + \frac{2x \sin(nx)}{n^2} + \frac{2 \cos nx}{n^3} \right]_{-\pi}^{\pi}$$

$$b_n = \frac{1}{n\pi} \left[-x^2 \cos(nx) + 2x \frac{\sin(nx)}{n} + \frac{2 \cos nx}{n^2} \right]_{-\pi}^{\pi}$$

$$b_n = \frac{1}{n\pi} \left[-\pi^2 \cos(n\pi) + 2\pi \frac{\sin(n\pi)}{n} + \frac{2 \cos(n\pi)}{n^2} - \left(-(-\pi)^2 \cos(-n\pi) + 2(-\pi) \frac{\sin(-n\pi)}{n} + \frac{2 \cos(-n\pi)}{n^2} \right) \right]$$

$$b_n = \frac{1}{n\pi} \left[-\pi^2 \cos(n\pi) + 0 + \frac{2 \cos(n\pi)}{n^2} + \pi^2 \cos(-n\pi) - 2 \cos(-n\pi) \right]$$

$$b_n = \frac{1}{n\pi} \left[-\pi^2 \cos(n\pi) + \frac{2 \cos(n\pi)}{n^2} + \pi^2 \cos(-n\pi) - \frac{2 \cos(-n\pi)}{n^2} \right]$$

$$b_n = \frac{-\pi \cos(n\pi)}{n} + \frac{2 \cos(n\pi)}{n^3 \pi} + \frac{\pi \cos(-n\pi)}{n} - \frac{2 \cos(-n\pi)}{n^3 \pi}$$

$$b_n = \cos(n\pi) \left[\frac{-\pi}{n} + \frac{2}{n^3 \pi} \right] + \cos(-n\pi) \left[\frac{\pi}{n} - \frac{2}{n^3 \pi} \right]$$

$$b_n = \cos(n\pi) \left[-\frac{\pi}{n} + \frac{2}{n^3\pi} \right] + \cos(-n\pi) \left[\frac{\pi}{n} - \frac{2}{n^3\pi} \right]$$

$n=1$

$$b_1 = \cos(\pi) \left[-\frac{\pi}{1} + \frac{2}{1^3\pi} \right] + \cos(-\pi) \left[\frac{\pi}{1} - \frac{2}{1^3\pi} \right]$$

$$b_1 = \cancel{\pi} - \frac{2}{\cancel{\pi}} - \cancel{\pi} + \frac{2}{\cancel{\pi}}$$

$$b_1 = 0$$

$$b_2 = 0$$

$$b_3 = 0$$

$$b_4 = 0$$

$$b_5 = 0$$

$$f(x) = \frac{1}{2} \left(\frac{2\pi^2}{3} \right) + (-4) + 1 + \left(-\frac{4}{9} \right) + \frac{1}{4} + \left(-\frac{4}{25} \right) + 0$$

$$f(x) = \frac{\pi^2}{3} - 4 \cos x + \cos 2x - \frac{4}{9} \cos 3x + \frac{1}{4} \cos 4x$$

$$- \frac{4}{25} \cos 5x$$

fourier sin series

Q: $f(x) = k \quad (0 < x < L)$

Sol:

$$\text{let } k = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

integrate

$$b_n = \frac{2k}{L} \int_0^L \sin\left(\frac{n\pi x}{L}\right) dx$$

$$b_n = \frac{2k}{L} \left[\frac{-\cos\left(\frac{n\pi x}{L}\right)}{\frac{n\pi}{L}} \right]_0^L$$

$$b_n = \frac{-2k}{n\pi} \left[\cos\left(\frac{n\pi x}{L}\right) \right]_0^L$$

$$b_n = \frac{-2k}{n\pi} \left[\cos\left(\frac{n\pi x}{L}\right) \right]_0^L$$

$$b_n = \frac{-2k}{n\pi} \left[\cos\left(\frac{n\pi L}{L}\right) - \cos\left(\frac{n\pi \cdot 0}{L}\right) \right]$$

$$b_n = \frac{-2k}{n\pi} \left[\cos(n\pi) - \cos(0) \right]$$

$$b_n = \frac{-2k}{n\pi} \left((-1)^n - 1 \right)$$

$$n=1$$

$$b_1 = \frac{-2k}{\pi} ((-1)^1 - 1) = \frac{-2k}{\pi} (-2) = \frac{4k}{\pi}$$

$$n=2$$

$$b_2 = \frac{-2k}{2\pi} (1-1) = 0$$

$$n=3$$

$$b_3 = \frac{-2k}{3\pi} (-1-1) = \frac{4k}{3\pi}$$

$$n=4$$

$$b_4 = \frac{-2k}{4\pi} (1-1) = 0$$

$$n=5$$

$$b_5 = \frac{-2k}{5\pi} (-1-1) = \frac{4k}{5\pi}$$

$$K = \frac{4k}{\pi} \sin\left(\frac{\pi x}{L}\right) + \frac{4k}{3\pi} \sin\left(\frac{3\pi x}{L}\right) + 0 + \frac{4k}{5\pi} \sin\left(\frac{5\pi x}{L}\right)$$

$$K = \frac{4k}{\pi} \left(\sin\left(\frac{\pi x}{L}\right) + \sin\left(\frac{3\pi x}{L}\right) + \sin\left(\frac{5\pi x}{L}\right) \right)$$



Ex - 2.4

Q: $3y'' + 10y' + 3y = x^2 + \sin x$

Sol:

Let $y'' = m^2$ $y' = m$ $y = 1$

Auxiliary eq

$$3m^2 + 10m + 3 = 0$$

$$3m^2 + 9m + m + 3 = 0$$

$$3m(m+3) + 1(m+3) = 0$$

$$(3m+1)(m+3) = 0$$

$$3m+1=0$$

$$m+3=0$$

$$3m = -1$$

$$m = -3$$

$$m = -\frac{1}{3}$$

$$y_h = C_1 e^{-3x} + C_2 e^{-\frac{1}{3}x}$$

$$y_p = K_2 x^2 + K_1 x + K_0$$

$$\sin x = A \cos x + B \sin x$$

$$\cos x = A \cos x + B \sin x$$

$$y_p = K_2 x^2 + K_1 x + K_0 + A \cos x + B \sin x$$

$$y'_p = 2K_2 x + K_1 - A \sin x + B \cos x$$

$$y''_p = 2K_2 - A \cos x - B \sin x$$

put in a

$$3(2k_2 - A\cos x - B\sin x) + 10(2k_2x + k_1 - A\sin x + B\cos x) + 3(k_2x^2 + k_1x + k_0 + A\cos x + B\sin x) = 0$$

ambarella

$$6k_2 - 3A\cos x - 3B\sin x + 20k_2x + 10k_1 - 10A\sin x + 10B\cos x + 3k_2x^2 + 3k_1x + 3k_0 + 3A\cos x + 3B\sin x = 0$$

$$3k_2x^2 + 20k_2x + 3k_1x + 10k_1 + 6k_2 + 3k_0 - 10A\sin x + 10B\cos x$$

comparing the coefficients of x^2

$$3k_2 = 1$$

$$k_2 = \frac{1}{3} \quad \text{compare coeff. of } x$$

$$20k_2 + 3k_1 = 0$$

$$20\left(\frac{1}{3}\right) + 3k_1 = 0$$

$$3k_1 = -\frac{20}{3}$$

$$k_1 = -\frac{20}{9}$$

compare constants

$$6k_2 + 10k_1 + 3k_0 = 0$$

$$2\left(\frac{1}{3}\right) + 10\left(-\frac{20}{9}\right) + 3k_0 = 0$$

$$2 - \frac{200}{9} + 3k_0 = 0$$

$$20k_2 + 3k_1 = 0$$

$$6k_2 + 3k_0 = 0$$

$$2\left(\frac{1}{3}\right) + 3k_0 = 0$$

$$3 + 3k_0 = 0$$

$$3k_0 = -3$$

$$k_0 = -1$$

$$3k_0 = \frac{200}{1} - 2$$

$$3k_0 = \frac{182}{1}$$

$$k_0 = \frac{182}{27}$$

compute coef of cos

$$-3A + 10B + 3A = 0$$

$$10B = 0$$

$$B = 0$$

compute coef of sin

$$-3B + 10A + 3B = 1$$

$$10A = 1$$

$$A = \frac{1}{10}$$

put k_2, k_1, k_0, A & B in y_p

$$y_p = \frac{1}{3}x^2 + \frac{-20}{9}x + \frac{182}{27} - \frac{1}{10}\cos x + 0$$

$$y = y_H + y_p$$

$$y = c_1 e^{-3x} + c_2 e^{-\frac{1}{3}x} + \frac{1}{3}x^2 - \frac{20}{9}x + \frac{182}{27} - \frac{1}{10}\cos x$$

2.10

Q3: $y'' - 2y' + y = 12e^x/x^3$

Sol: $y'' + P(x)y' + Q(x)y = r(x)$

$$y'' - 2y' + y = 0$$

$$m^2 - 2m + 1 = 0$$

$$m^2 - m - m + 1 = 0$$

$$m(m-1) - 1(m-1) = 0$$

$$(m-1)(m-1) = 0$$

$$m-1 = 0$$

$$m = 1$$

roots are equal

$$y_H = (c_1 + c_2 x) e^x$$

$$y_H = c_1 e^x + c_2 x e^x$$

let

$$y_1 = e^x$$

$$y_1' = e^x$$

$$y_2 = x e^x$$

$$y_2' = x e^x + e^x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & x e^x \\ e^x & x e^x + e^x \end{vmatrix}$$

$$W = e^x(x e^x + e^x) - e^x(x e^x)$$

$$W = \cancel{x e^{2x}} + e^{2x} - \cancel{x e^{2x}}$$

$$W = e^{2x}$$

$$\begin{aligned}
y_p &= -y_1 \int \frac{y_2 r(x) dx}{w} + y_2 \int \frac{y_1 r(x) dx}{w} \\
&= -e^x \int \frac{x e^x 12 e^x x^{-3}}{e^{3x}} dx + x e^x \int \frac{e^x 12 e^x x^{-3}}{e^{2x}} dx \\
&= -12 e^x \int x^{-2} dx + 12 x e^x \int x^{-3} dx \\
&= -12 e^x \frac{x^{-1}}{-1} + \frac{6}{2} x e^x \frac{x^{-2}}{-2} + C \\
&= 12 e^x x^{-1} - \frac{6}{2} x e^x x^{-2} + C \\
&= 6 e^x (2 x^{-1} - x^{-1}) + C \\
&= 6 e^x (x^{-1}) + C \\
&= \frac{6 e^x}{x} + C
\end{aligned}$$

$$y = y_h + y_p$$

$$y = c_1 e^x + c_2 x e^x + \frac{6 e^x}{x} + C$$

$$\text{Q9: } y'' - 2y' + y = e^x \sin x$$

Sol:

$$y'' - 2y' + y = 0$$

$$m^2 - 2m + 1 = 0$$

$$m^2 - m - m + 1 = 0$$

$$m(m-1) - 1(m-1) = 0$$

$$(m-1)(m-1) = 0$$

$$m=1$$

$$m=1$$

roots are same

$$y_H = (c_1 + c_2 x) e^x$$

$$y_H = c_1 e^x + c_2 x e^x$$

Let $y_1 = e^x$
 $y_1' = e^x$

$y_2 = x e^x$
 $y_2' = x e^x + e^x$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & x e^x \\ e^x & x e^x + e^x \end{vmatrix}$$

$$W = e^x (x e^x + e^x) - e^x (x e^x)$$

$$W = x e^{2x} + e^{2x} - \cancel{e^{2x} x}$$

$$W = e^{2x}$$

$$y_p = -y_1 \int \frac{y_2 r(x)}{w} dx + y_2 \int \frac{y_1 r(x)}{w} dx$$

$$y_p = -e^x \int \frac{x e^x e^x \sin x}{e^{2x}} dx + x e^x \int \frac{e^x e^x \sin x}{e^{2x}} dx$$

$$y_p = -e^x \int \frac{x e^{2x} \sin x}{e^{2x}} dx + x e^x \int \frac{e^{2x} \sin x}{e^{2x}} dx$$

$$y_p = -e^x \int x \sin x dx + x e^x \int \sin x dx$$

$$y_p = -e^x (-x \cos x + \sin x) + x e^x (-\cos x) + C$$

x	$\sin x$
1	$-\cos x$
0	$-\sin x$

$$y_p = \cancel{x e^x \cos x} - e^x \sin x - \cancel{x e^x \cos x} + C$$

$$y_p = -e^x \sin x + C$$

$$y = y_H + y_p$$

$$y = c_1 e^x + c_2 x e^x - e^x \sin x + C$$

$$\textcircled{10} : y'' + 6y' + 9y = 8e^{-3x} / (x^2 + 1)$$

$$m^2 + 6m + 9 = 0$$

$$m^2 + 3m + 3m + 9 = 0$$

$$m(m+3) + 3(m+3) = 0$$

$$(m+3)(m+3) = 0$$

$$m = -3$$

$$m = -3$$

roots are same

$$y_h = (c_1 + c_2 x) e^{-3x}$$

$$y_h = c_1 e^{-3x} + c_2 x e^{-3x}$$

$$\text{Let } y_1 = e^{-3x}$$

$$y_1' = -3e^{-3x}$$

$$y_2 = x e^{-3x}$$

$$y_2' = -3x e^{-3x} + e^{-3x}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-3x} & x e^{-3x} \\ -3e^{-3x} & -3x e^{-3x} + e^{-3x} \end{vmatrix}$$

$$W = e^{-3x} (-3x e^{-3x} + e^{-3x}) - x e^{-3x} (-3e^{-3x})$$

$$W = -3x e^{-6x} + e^{-6x} + 3x e^{-6x}$$

$$W = e^{-6x}$$

★
answer
check

$$\begin{aligned}y_p &= -y_1 \int \frac{y_2 \delta(x)}{w} dx + y_2 \int \frac{y_1 \delta(x)}{w} dx \\&= -e^{-3x} \int \frac{x e^{-3x} 8e^{-3x}}{e^{-6x}(x^2+1)} dx + x e^{-3x} \int \frac{e^{-3x} 8e^{-3x}}{e^{-6x}(x^2+1)} dx \\&= -e^{-3x} \int \frac{8x e^{-6x}}{e^{-6x}(x^2+1)} dx + x e^{-3x} \int \frac{8e^{-6x}}{e^{-6x}(x^2+1)} dx \\&= -8e^{-3x} \int \frac{x}{x^2+1} dx + 8x e^{-3x} \int \frac{1}{x^2+1} dx \\&= -4e^{-3x} \int \frac{2x}{x^2+1} dx + 8x e^{-3x} \int (x^2+1)^{-1} dx\end{aligned}$$

$$y_p = -4e^{-3x} \ln(x^2+1) + 8x e^{-3x} \tan^{-1}(x) + C$$

$$y = c_1 e^{-3x} + c_2 x e^{-3x} - 4e^{-3x} \ln(x^2+1) + 8x e^{-3x} \tan^{-1}(x) + C$$

Cl 3

Chp - 11

Q1: $u_x + u_y = 0$

Soln

Let $u(x, y) = f(x) \times g(y)$ — (1) key

diff w.r.t. (x)

$$u_x = f'(x) g(y)$$

diff w.r.t. (y)

$$u_y = f(x) g'(y)$$

put in eqn

$$f'(x) g(y) + f(x) g'(y) = 0$$

$$f'(x) g(y) = -f(x) g'(y)$$

$$\frac{f'(x)}{f(x)} = -\frac{g'(y)}{g(y)} = k$$

$$\frac{f'(x)}{f(x)} = k$$

integrate

$$\int \frac{f'(x)}{f(x)} dx = \int k dx$$

$$\ln(f(x)) = kx + c$$

$$e^{\ln(f(x))} = e^{kx+c}$$

$$f(x) = e^{kx+c}$$

$$-\frac{g'(y)}{g(y)} = k$$

$$\int \frac{g'(y)}{g(y)} = -k \int dx$$

$$\ln(g(y)) = -kx + c$$

$$e^{\ln(g(y))} = e^{-kx+c}$$

$$g(y) = e^{-kx+c}$$

put $f(x)$ & $g(y)$ in keypoint

$$\begin{aligned}u(x, y) &= f(x) g(y) \\&= e^{kx+c} (e^{-ky+c_2}) \\&= e^{kx+c} e^{-ky+c_2} \cancel{e^{kx+c}} \\&= \cancel{e^c} e^{kx+c_1 - ky+c_2} \\&= e^{k(x-y) + c} \\&= e e^{k(x-y)} \\&= e c e^{k(x-y)}\end{aligned}$$

$\Rightarrow$

Q Q3: $ux + uy = 2(x+y) u$

So let $u(x,y) = f(x) g(y)$ key point

$$ux = f'(x) g(y)$$

$$uy = f(x) g'(y)$$

put values

$$f'(x) g(y) + f(x) g'(y) = 2(x+y) f(x) g(y)$$

divide $f(x) g(y)$

$$\frac{f'(x)}{f(x)} + \frac{g'(y)}{g(y)} = 2(x+y)$$

$$\frac{f'(x)}{f(x)} + \frac{g'(y)}{g(y)} = 2x + 2y$$

$$\frac{f'(x)}{f(x)} - 2x = -\frac{g'(y)}{g(y)} + 2y = k$$

a $\frac{f'(x)}{f(x)} - 2x = k$

$$\frac{g'(y)}{g(y)} - 2y = -k$$

$$\ln(f(x)) - \frac{x^2}{2} = kx + C_1$$

$$\ln(g(y)) - \frac{y^2}{2} = -ky + C_2$$

$$f(x) = \cancel{e^{x^2}}$$

$$\ln(f(x)) = x^2 + kx + c_1$$

$$f(x) = e^{x^2 + kx + c_1}$$

$$\ln(g(y)) = y^2 - ky + c_2$$

$$g(y) = e^{y^2 - ky + c_2}$$

put values in key point

$$\begin{aligned} u(x,y) &= e^{x^2 + kx + c_1} \cdot e^{y^2 - ky + c_2} \\ &= e^{x^2 + kx + c_1 + y^2 - ky + c_2} \\ &= e^{x^2 + y^2 + k(x-y) + c} \\ &= C e^{x^2 + y^2 + k(x-y)} \end{aligned}$$



11.4

Q: $u_{xy} - u_{yz} = 0$

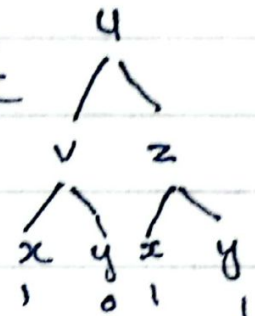
$v = x$ $z = x + y$

soln

$$u_x = \frac{\partial u}{\partial v} \times \frac{\partial v}{\partial x} + \frac{\partial u}{\partial z} \times \frac{\partial z}{\partial x}$$

$u_x = u_v(1) + u_z(1)$

$u_x = u_v + u_z$



$$u_{xy} = u_v + u_z \left[\frac{\partial u}{\partial v} \frac{\partial v}{\partial y} + \frac{\partial u}{\partial z} \frac{\partial z}{\partial y} \right]$$

$$u_{xy} = (u_v + u_z) \left[\frac{\partial u}{\partial v} (0) + \frac{\partial u}{\partial z} (1) \right]$$

$u_{xy} = (u_v + u_z) (0 + u_z)$

$u_{xy} = (u_v + u_z) u_z$

$u_{xy} = u_{vz} + u_{zz}$

$$u_y = \frac{\partial u}{\partial v} \frac{\partial v}{\partial y} + \frac{\partial u}{\partial z} \frac{\partial z}{\partial y}$$

$u_y = 0 + u_z$

$u_y = u_z$

$$u_{yy} = u_z \left[\frac{\partial u}{\partial v} \frac{\partial v}{\partial y} + \frac{\partial u}{\partial z} \frac{\partial z}{\partial y} \right]$$

$$u_{yy} = u_z [0 + u_z]$$

$$u_{yy} = u_z (u_z)$$

$$u_{yy} = u_{zz}$$

$$u_{vz} + u_{zz} - u_{zz} = 0$$

$$u_{vz} = 0$$



non homogeneous

$$Q: \frac{dy}{dx} = \frac{2y-x+5}{2x-y-4} \quad \text{--- (1)}$$

Sol:

$$-x \rightarrow a_1$$

$$2y \rightarrow b_1$$

$$2x \rightarrow a_2$$

$$-y \rightarrow b_2$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$\frac{-1}{2} \neq \frac{2}{-1}$$

$$\text{let } x = x+h$$

$$y = y+k$$

$$\frac{dy}{dx} = \frac{2(y+k) - (x+h) + 5}{2(x+h) - (y+k) - 4}$$

$$\frac{dy}{dx} = \frac{2y+2k-x-h+5}{2x+2h-y-k-4}$$

$$\frac{dy}{dx} = \frac{2y-x+2k-h+5}{2x-y+2h-k-4} \quad \text{--- (i)}$$

$$\text{let } 2k - h + 5 = 0 \quad \text{--- (a)}$$

$$\text{and } 2h - k - 4 = 0 \quad \text{--- (b)}$$

multiply (b) by 2 and add in (a)

$$2k - h + 5 = 0$$

$$\underline{-2k + 4h - 8 = 0}$$

$$3h - 3 = 0$$

$$3h = +3$$

$$h = +1$$

put $h = 1$ in (a)

$$2k - 1 + 5 = 0$$

$$2k + 4 = 0$$

$$2k = -4$$

$$k = -2$$

put h & k in (i)

$$\frac{dy}{dx} = \frac{2y - x + 2(-2) - 1 + 5}{2x - y + 2(1) - (-2) - 4}$$

$$\frac{dy}{dx} = \frac{2y - x - 4 - 1 + 5}{2x - y + 2 + 2 - 4}$$

$$\frac{dy}{dx} = \frac{2y - x}{2x - y}$$

homogeneous eq

$$\frac{dy}{dx} = \frac{2y-x}{2x-y}$$

$$\text{let } y = ux$$

$$y' = u + x \frac{du}{dx}$$

$$u + x \frac{du}{dx} = \frac{2(ux) - x}{2x - ux}$$

$$u + x \frac{du}{dx} = \frac{x(2u-1)}{x(2-u)}$$

$$u + x \frac{du}{dx} = \frac{2u-1}{2-u}$$

$$x \frac{du}{dx} = \frac{2u-1}{2-u} - u$$

$$x \frac{du}{dx} = \frac{2u-1 - u(2-u)}{2-u}$$

$$x \frac{du}{dx} = \frac{2u-1-2u+u^2}{2-u}$$

$$x \frac{du}{dx} = \frac{u^2-1}{2-u}$$

$$\frac{2-u}{u^2-1} du = \frac{1}{x} dx$$

integrate

$$\int \frac{2-u}{u^2-1} du = \int \frac{1}{x} dx$$

$$\int \frac{2}{u^2-1} du = \int \frac{u}{u^2-1} du = \ln x + C$$

$$2 \int \frac{1}{u^2-1} du = \frac{1}{2} \int \frac{2u}{u^2-1} du = \ln x + C$$

$$2 \left(\int \frac{1}{(u+1)(u-1)} du \right) = \frac{1}{2} \ln(u^2-1) = \ln x + C$$

$$2 \left(\int \frac{A}{u+1} + \frac{B}{u-1} du \right) = \frac{1}{2} \ln(u^2-1) = \ln x + C$$

$$2 \left(\int \frac{A}{u+1} du + \int \frac{B}{u-1} du \right) = \frac{1}{2} \ln(u^2-1) = \ln x + C$$

$$2 \left(A \int \frac{1}{u+1} du + B \int \frac{1}{u-1} du \right) = \frac{1}{2} \ln(u^2-1) = \ln x + C$$

$$2 \left(A \ln \frac{(u+1)}{u} + B \ln \frac{(u-1)}{u} \right) = \frac{1}{2} \ln(u^2-1) = \ln x + C$$

$$2A \ln \frac{(u+1)}{u} + 2B \ln \frac{(u-1)}{u} = \frac{1}{2} \ln(u^2-1) = \ln x + C$$

practice
we
need

$$\text{as } u = \frac{y}{x}$$

$$2A \ln\left(\frac{y}{x} + 1\right) + 2B \ln\left(\frac{y}{x} - 1\right) - \frac{1}{2} \ln\left(\frac{y^2}{x^2} - 1\right) = \ln x + \ln$$

$$\ln\left(\frac{y}{x} + 1\right)^{2A} + \ln\left(\frac{y}{x} - 1\right)^{2B} - \ln\left(\frac{y^2}{x^2} - 1\right)^{1/2} = \ln x + \ln$$

taking e on b.s

$$\left(\frac{y}{x} + 1\right)^{2A} + \left(\frac{y}{x} - 1\right)^{2B} - \left(\frac{y^2}{x^2} - 1\right)^{1/2} = x + c$$

